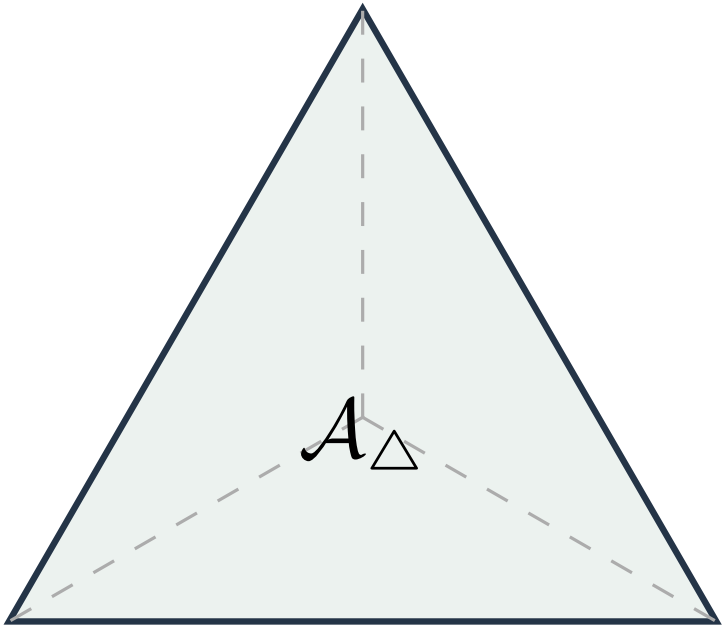

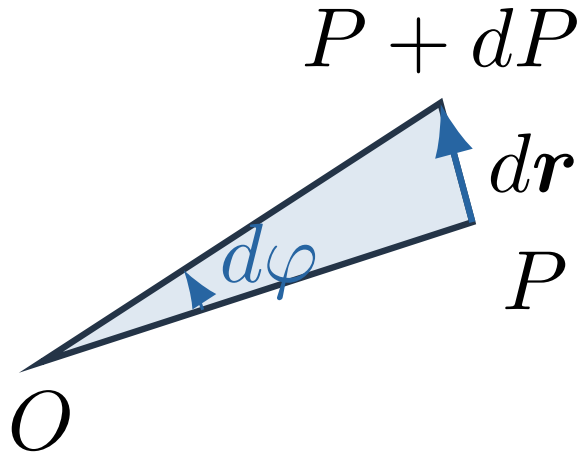

$$d\mathcal{A} = \tfrac{1}{2}\rho^2 d\varphi$$